

Movies Revenue and Ratings Prediction: An Analysis of TMDB Data from 2010 to 2024

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1 Introduction

Understanding what makes a movie commercially and critically successful is a long-standing question in both the film industry and in data science. In this project, we study 3,532 movies released between 2010 and 2024 collected from The Movie Database (TMDB), and we formulate two supervised learning tasks using only *pre-release* attributes:

- **Task A (Regression):** Predict the log-transformed box-office revenue, `log_revenue`.
- **Task B (Classification):** Predict `high_rating`, a binary indicator of whether a film receives a TMDB audience vote average of at least 7.0.

We deliberately restrict the feature set to only the information that is known before release, which includes budget, runtime, genres, production companies and countries, temporal release features, and cast popularity derived from prior work. In this way, the resulting models have practical value for planning and marketing decisions.

Across four model families (a naive baseline, a regularized linear model, a random forest, and a gradient boosting machine), we find that commercial success is substantially more predictable than critical success. The best regression model (Ridge) achieves $R^2 = 0.538$ on a held-out test set, driven overwhelmingly by the film’s budget. In contrast, the best classifier (Gradient Boosting) attains ROC-AUC = 0.754 on high-rating prediction, with no single feature dominating and with budget surprisingly exhibiting a *negative* association with high ratings. Threshold tuning on the validation year raises test F_1 from 0.577 to 0.619. These results, along with a qualitative error analysis, suggest that structured pre-release metadata captures much of the variance in revenue but only a modest portion of the variance in perceived quality.

2 Data Description

2.1 Data source

Data were collected from the TMDB API. We used the `/discover/movie` endpoint to enumerate titles released between 2010 and 2024 and the `/movie/{id}` endpoint to retrieve metadata, crew, cast, and keyword fields for each film.

2.2 Cleaning and filtering

The raw feed contains many fragmentary entries (zero budget, missing revenue, adult content, very short runtimes, or almost no votes), which would severely bias the modelling tasks. We therefore retain only movies satisfying all of the following:

- released status, non-adult;
- budget > 0 and revenue > 0 ;
- runtime > 40 minutes;
- TMDb vote count ≥ 50 .

These filters yield a cleaned dataset of 3,532 films.

2.3 Feature engineering

We construct four families of features (77 in total):

- **Production:** `log_budget`, `runtime`, multi-hot indicators for the top-20 production companies and top-15 production countries, plus aggregate indicators (`is_us_production`, `is_international_coproduction`, `num_production_companies`, `num_production_countries`).
- **Temporal:** `release_year`, `release_month`, `release_quarter`, `release_dayofweek`, and binary flags `is_summer_release`, `is_holiday_season_release`.
- **Content:** multi-hot genre indicators across all unique genres; lengths of `overview` and `tagline`; `has_tagline`; `num_keywords`; `num_genres`.
- **Talent:** `director_popularity` and several cast popularity summaries (`top_cast_popularity`, `top3_cast_avg_pop`, `top5_cast_avg_pop`, `max_cast_popularity`, `num_cast`).

2.4 Train/validation/test split

Because movies released in one year are informative about films released in the same or adjacent years, a random split would leak future information. We therefore use a *time-based split*:

Split	Year range	n	High-rating ratio
Train	≤ 2021	2,976	0.278
Validation	2022	166	0.398
Test	2023–2024	390	0.341

Table 1: Time-based split used for both tasks.

The high-rating prior is approximately 28–40% across splits, i.e. the classification target is only moderately imbalanced.

3 Models and Methods

3.1 Problem formulations

For Task A we minimize squared error on $y = \log(\text{revenue})$, reported together with mean absolute error and coefficient of determination R^2 . For Task B we model $P(\text{vote_average} \geq 7 \mid \mathbf{x})$ and report ROC-AUC, accuracy, F_1 , and the predicted positive rate.

3.2 Model families

For both tasks we compare four families:

- **Baseline:** Two Dummy models are implemented. `strategy="mean"` for regression and `strategy="stratified"` for classification. These calibrate how much signal actually exists beyond the class prior or mean target.
- **Linear model:** Ridge regression ($\alpha = 1$) for Task A and ℓ_2 -regularized logistic regression ($C = 1$, `class_weight="balanced"`, `max_iter = 2000`) for Task B. Both are embedded in a Pipeline with `StandardScaler` so that the reported coefficients can be compared on a common scale.
- **Random Forest:** 400 trees, unconstrained depth, `min_samples_leaf=2`. For classification we again use `class_weight="balanced"`.
- **Gradient Boosting:** 400 trees with `max_depth=3` and `learning_rate=0.05`, the standard scikit-learn implementation.

All models are fit on the training partition, compared on the 2022 validation partition, and finally evaluated once on the 2023–2024 test partition to simulate genuine forward-looking prediction.

3.3 Threshold tuning

The default 0.5 decision threshold is rarely F_1 -optimal when the positive rate is moderate. We therefore sweep thresholds $t \in [0.05, 0.95]$ on the validation set, record $F_1(t)$, and re-evaluate the chosen threshold on the test set.

4 Results and Interpretation

4.1 Task A: revenue regression

Table 2 shows the test-set metrics for each regression model.

Model	Val MAE	Val RMSE	Val R^2	Test RMSE	Test R^2
Ridge	1.089	1.482	0.535	1.256	0.538
Gradient Boosting	1.259	1.615	0.447	1.469	0.368
Random Forest	1.272	1.690	0.395	1.527	0.317
Baseline (mean)	1.764	2.186	−0.012	1.853	−0.005

Table 2: Regression models on the 2023–2024 test set.

Ridge is clearly the strongest model, cutting test RMSE by $\approx 32\%$ relative to the mean baseline and explaining roughly 54% of the variance in log-revenue. The two non-linear models lag substantially. This inversion is perhaps surprising, but it is consistent with two well-known facts: (i) once budget is on a log-scale, its relationship to log-revenue is close to linear, so the added flexibility of trees yields little benefit; and (ii) with $\sim 3k$ training rows and ~ 80 features, deep tree ensembles are more vulnerable to variance than a well-regularized linear model.

Figure 1 confirms this. The predicted-vs-actual scatter lies tightly around the identity line except in the lower tail, and residuals are roughly centered on zero with no systematic structure.

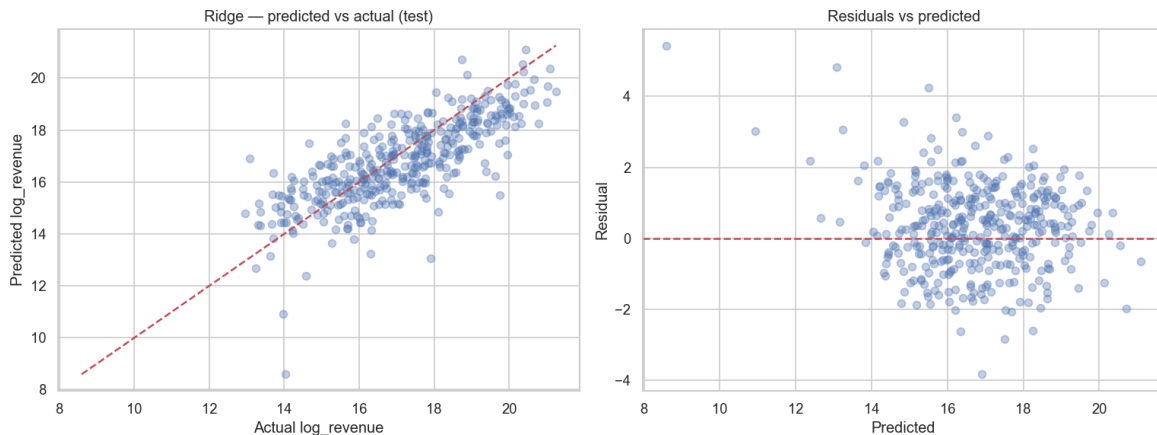


Figure 1: Regression diagnostics on the 2023–2024 test set for the best model (Ridge). Left: predicted vs. actual `log_revenue`. Right: residuals vs. predicted values.

The standardized Ridge coefficients (Figure 2, right) place an order-of-magnitude weight on `log_budget` (0.86), followed by `num_keywords` (0.27), `top5_cast_avg_pop` (0.25), and a moderate negative seasonal effect on `release_month`. The Random Forest importance ranking (Figure 2, left) agrees: `log_budget` alone accounts for 54% of the total importance, with the rest shared between `release_year`, `num_keywords`, `director_popularity`, and `runtime`. In short, how much a studio invests remains by far the most informative pre-release signal about box-office outcome.

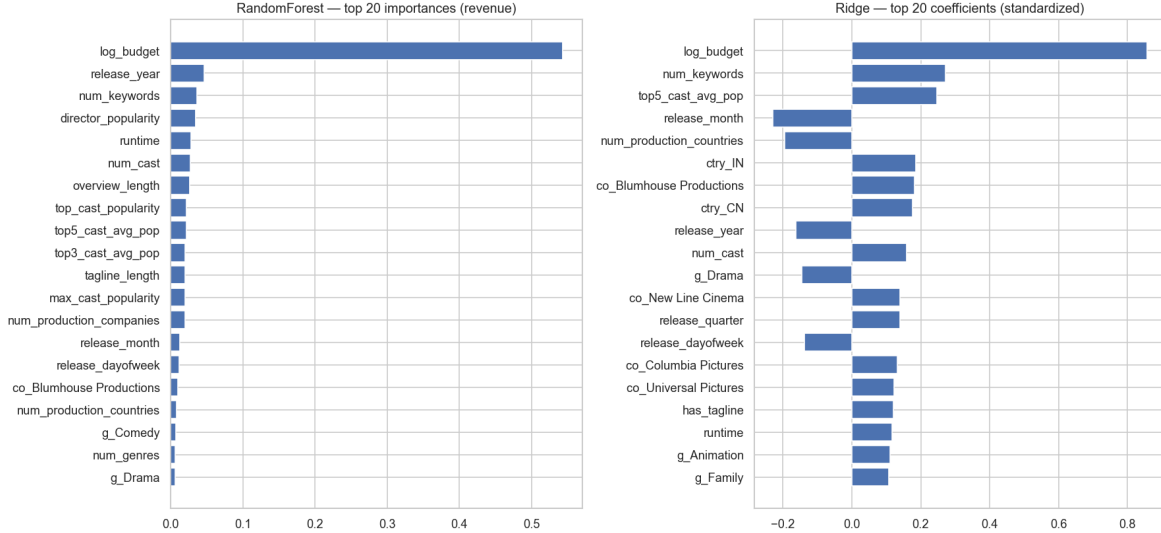


Figure 2: Feature importances for revenue prediction. Left: Random Forest Gini importance (top-20). Right: standardized Ridge coefficients (top-20).

Regression error analysis. Inspecting the largest absolute residuals on the test set reveals two recurring failure modes. First, *sleeper hits* with modest budgets dramatically outperform their priors: *Sound of Freedom* (2023), *TAYLOR SWIFT | THE ERAS TOUR* (2023), and *How to Make Millions Before Grandma Dies* (2024) all under-predict by ≥ 2.6 log-units. Second, *genre-specific flops* (e.g. *The Kill Room*, *The Bricklayer*, *Hitpig!*) fool the model because their production features look conventional. These cases highlight an unavoidable limit of pre-release prediction: word-of-mouth and cultural moments are not observable in metadata.

4.2 Task B: high-rating classification

Results for the binary classification task are summarized in Table 3.

Model	Val AUC	Test AUC	Test F_1	Test acc.
Gradient Boosting	0.750	0.754	0.577	0.703
Random Forest	0.744	0.744	0.520	0.697
Logistic Regression	0.760	0.725	0.589	0.574
Baseline (stratified)	0.446	0.478	0.269	0.554

Table 3: Classification models on the 2023–2024 test set.

Three observations stand out. First, all three learned models beat the stratified baseline by a wide margin on AUC ($\Delta \approx 0.25$), so the task is clearly learnable. Second, the ceiling is modest even Gradient Boosting reaches only AUC = 0.754, far below the ~ 0.9 that is typical for revenue regression benchmarks. Third, the models trade off differently: the ℓ_2 -regularized logistic regression with balanced class weights maintains a high positive rate and a marginally better F_1 , whereas the tree-based models are more conservative with positive predictions and therefore achieve higher accuracy at a lower F_1 .

The ROC curve, precision–recall curve, and confusion matrix at the default 0.5 threshold are shown in Figure 3.

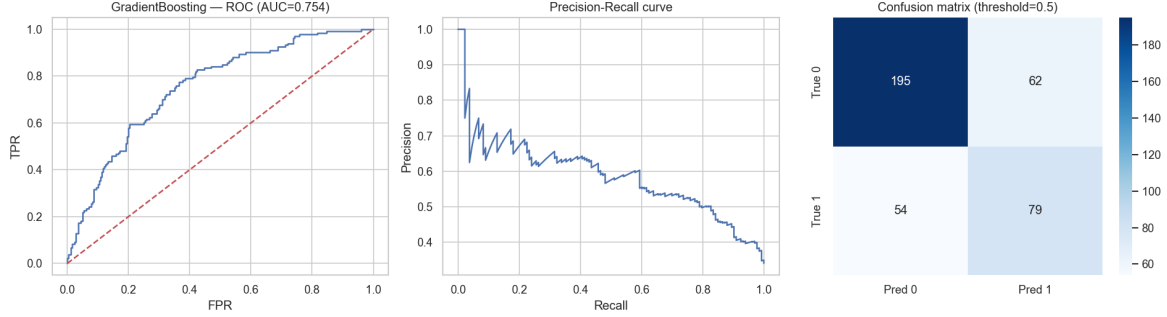


Figure 3: Classification diagnostics on the 2023–2024 test set for the best model (Gradient Boosting). Left to right: ROC curve, precision–recall curve, and confusion matrix at threshold 0.5.

Threshold tuning. Figure 4 plots F_1 on the validation set as a function of the decision threshold. The optimum is $t^* = 0.36$, well below 0.5, which raises the test F_1 from 0.577 to **0.619** while leaving AUC unchanged.

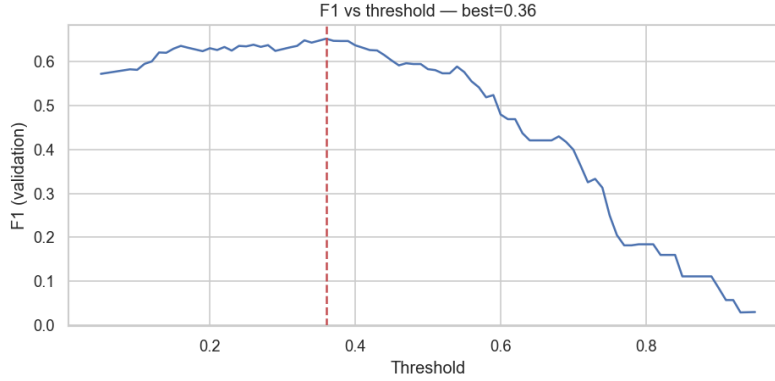


Figure 4: F_1 on the 2022 validation set as a function of the decision threshold; the vertical dashed line shows the chosen value $t^* = 0.36$.

Feature importances and a counter-intuitive finding. The top classification features (Figure 5) contrast sharply with those for regression. `runtime`, `num_keywords`, `director_popularity`, cast popularity summaries, and `overview_length` all matter; no single variable dominates, and budget’s importance is substantially reduced. More striking is the *sign* of `log_budget` in the logistic regression (-0.40): holding other features fixed, higher-budget movies are *less* likely to reach a 7.0 vote average. The Animation genre flag has a large positive coefficient, while Horror and Action carry negative weights. Taken together, the model learns that long, substantive, animation- or drama-leaning films with experienced directors are the most likely to be loved by audiences, whereas large-budget action vehicles frequently disappoint.

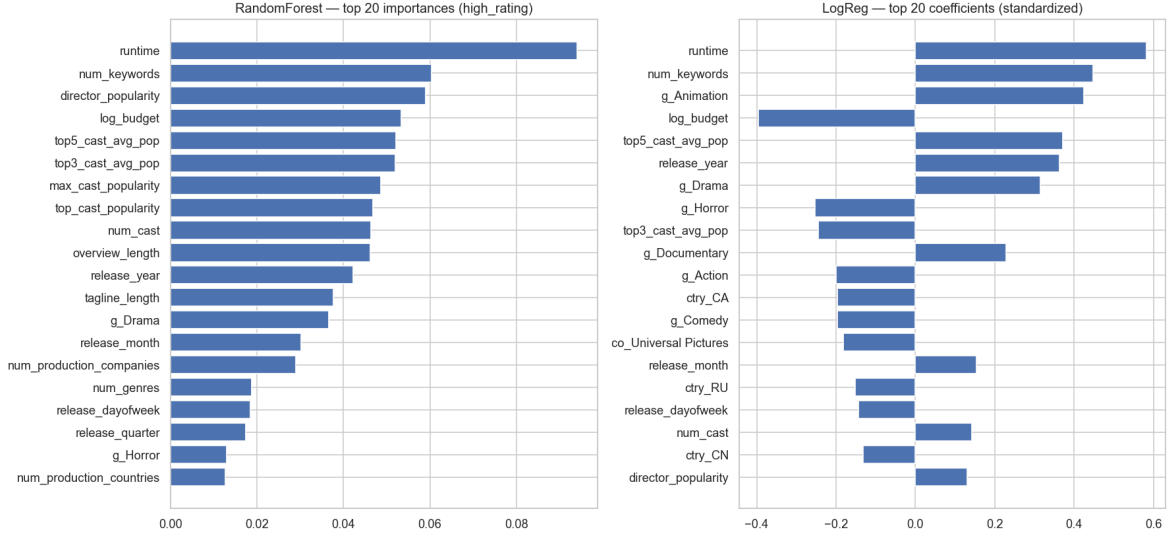


Figure 5: Feature importances for high-rating classification. Left: Random Forest Gini importance (top-20). Right: standardized logistic-regression coefficients (top-20).

Classification error analysis. The confident mistakes are especially illuminating. Among the high-probability false positives we find *Wicked* (2024, $\hat{p} = 0.95$, actual 6.91), *Wish* (2023, 6.22), *Megalopolis* (2024, 5.19), and *Kinds of Kindness* (2024, 6.50): each has the *pre-release fingerprint* of a prestige project (long runtime, big budget, experienced director, rich metadata) but underwhelmed audiences. Conversely, the most confident false negatives — *Five Nights at Freddy’s* (7.33), *Talk to Me* (7.14), *Speak No Evil* (7.13) — are mid-budget horror/thrillers whose genre coefficient is negative in the linear model but that turn out to be audience favourites. These errors are precisely the ones that pre-release features *cannot* anticipate: they depend on execution and cultural timing rather than on measurable inputs.

5 Conclusion and Next Steps

We built and compared baseline, linear, random forest, and gradient boosting models for two complementary pre-release prediction tasks on TMDB data from 2010–2024 under a strict time-based split. On the 2023–2024 test set, Ridge regression explains about 54% of the variance in log-revenue (test RMSE = 1.256), with budget as the overwhelming driver. On the high-rating task the best model is Gradient Boosting with ROC-AUC = 0.754 and, after validation-tuned threshold selection, $F_1 = 0.619$. The two tasks rely on largely different feature sets: commercial success is dominated by budget, whereas critical success is explained by runtime, keywords, director experience, and genre, with budget contributing *negatively*.

Several natural extensions follow from this analysis:

- **Richer text features.** We currently summarize the overview and tagline by their length. Sentence-level embeddings (e.g. sentence-transformers) or topic models would expose narrative content that is plausibly informative about audience reception.
- **Track-record features.** Director and cast popularity are point-in-time proxies. Rolling historical statistics (prior films’ average ratings, cumulative box-office, frequency of prior collaborations) would provide a much stronger notion of “experience” for both tasks.

- **Probability calibration.** The Gradient Boosting classifier is not calibrated; isotonic or Platt scaling on the validation set could meaningfully improve decisions when predicted probabilities are used downstream.
- **Heavy-tailed revenue.** Residual analysis suggests an asymmetric tail: sleeper hits are systematically under-predicted. Quantile regression, a Tweedie-family GLM, or dedicated models for the upper tail could complement the point estimates.
- **Budget–rating trade-off.** The negative association between budget and high ratings merits a dedicated causal or stratified analysis e.g. quantifying how budget affects ratings conditional on genre and franchise status.

Overall, the results support a nuanced conclusion: *pre-release metadata is a strong predictor of how much money a film will make but only a moderate predictor of how much audiences will like it.* Bridging that gap likely requires features that capture narrative, execution, and cultural context rather than purely structural attributes.